

TECHNICAL DOCUMENTATION: MUD PUMP SYSTEM

IDENTITY INFORMATION

****Company_Name:** Permian Basin Drilling Solutions LLC**

****Field_Engineer:** Marcus J. Whitmore**

****Field_Engineer_Id:** PBD-FE-4782**

****Document_Version:** 2.3.1**

****Machine_Name:** Triplex Mud Pump Assembly – Primary Circulation Unit**

****Machine_Id:** PBDS-TMP-2847-A**

****Installation_Date:** 2023-06-15**

****Number_Of_Units_In_Machine:** 3**

UNIT 1: PRIMARY TRIPLEX PUMP HEAD

****Unit_Name:** Triplex Pump Head – Stage 1 (High-Pressure Discharge)**

****Unit_Id:** PBDS-TPH-2847-A1**

****Manufacture_Date:** 2023-04-22**

****Manufacturer_Company_Name:** Weatherford International Drilling Equipment**

****Manufacturer_Company_Id:** WI-DEQ-7734**

****Inventory_Quantity:** 2 units in stock**

****Supplier_Information:** Weatherford Houston Distribution Center, Contract #WI-2023-4156, Lead Time: 14 days**

DOMAIN SPECIFICATIONS:

- ****Wellbore_Depth_Rating:** 15,000 feet TVD (True Vertical Depth)**

- ****Operating_Pressure_PSI:** 3,500 PSI nominal / 4,200 PSI maximum relief setting**

- ****Drilling_Fluid_Compatibility:** Water-based mud, synthetic-based mud, oil-based mud (OBM), weighted mud systems up to 18 ppg**

- ****Known_Error_Codes:** ERR-TPH-001 (Valve seat erosion), ERR-TPH-002 (Piston rod scoring), ERR-TPH-003 (Fluid bypass at discharge)**

- ****Replacement_Part_Number:** WI-TPH-3500-R2**

- ****Maintenance_Schedule:** 500-hour inspection interval; valve seat replacement every 2,000 operating hours; complete overhaul every 8,000 hours**

SECTION 1 – USE CASE

- Primary circulation pump responsible for continuous drilling fluid delivery to wellbore at controlled pressure and flow rate

- Maintains hydrostatic pressure balance and cuttings transport during drilling operations
- Operates in triplex configuration (three pistons) for smooth, consistent flow delivery
- Supports mud weight management and well control procedures per API RP 54 Section 5.2
- Integrates with automated pressure monitoring and relief valve systems for safety compliance

SECTION 2 – FAILURE CAUSES

- ****ERR-TPH-001 (Valve Seat Erosion)**** – Frequency: 34% of failures; caused by abrasive particle impingement and cavitation erosion in high-velocity discharge zones
- ****ERR-TPH-002 (Piston Rod Scoring)**** – Frequency: 28% of failures; results from inadequate lubrication film, contaminated mud, or misalignment of rod guides
- ****ERR-TPH-003 (Fluid Bypass at Discharge)**** – Frequency: 22% of failures; seal degradation from thermal cycling and incompatible fluid chemistry
- ****Bearing Cage Fracture**** – Frequency: 10% of failures; fatigue failure under sustained high-pressure cycling
- ****Suction Valve Blockage**** – Frequency: 6% of failures; mud solids accumulation in inlet manifold

SECTION 3 – SOLUTIONS

- ****ERR-TPH-001 Mitigation:**** Replace valve seat assembly per WI-TPH-3500-R2 specification; implement mud conditioning system (shale shakers, centrifuges) to reduce particle size per API RP 13C; inspect discharge manifold for cavitation damage
- ****ERR-TPH-002 Mitigation:**** Perform rod guide alignment check using dial indicators (tolerance ± 0.005 "); flush lubrication system with compatible hydraulic fluid; replace piston rod if scoring depth exceeds 0.015"
- ****ERR-TPH-003 Mitigation:**** Replace primary and secondary seals per OSHA 1910.119 pressure equipment standards; verify fluid compatibility with elastomer materials; monitor discharge temperature (max 180°F); implement thermal management protocol
- ****Bearing Cage Fracture:**** Replace bearing assembly; reduce operating pressure by 10% during next 100 hours; schedule metallurgical analysis of failed bearing
- ****Suction Valve Blockage:**** Clean inlet manifold; install 200-micron suction strainer; increase mud tank circulation frequency

SECTION 4 – INVENTORY DATA

- ****Current Stock:**** 2 complete pump head assemblies; 8 valve seat kits; 6 piston rod assemblies; 12 seal kits
- ****Reorder Threshold:**** Maintain minimum 1 complete unit and 4 valve seat kits on-site; trigger purchase order when stock reaches threshold
- ****Supplier:**** Weatherford Houston Distribution Center; secondary supplier: Smith Services Equipment (Contract #SSE-2023-1847)
- ****Lead Time:**** 14 days standard; 7 days expedited (25% surcharge); emergency 48-hour delivery available for critical components

UNIT 2: SUCTION MANIFOLD ASSEMBLY

****Unit_Name:** Suction Manifold with Inlet Strainer and Check Valve**

****Unit_Id:** PBDS-SMA-2847-A2**

****Manufacture_Date:** 2023-05-08**

****Manufacturer_Company_Name:** Halliburton Drilling Fluids Equipment Division**

****Manufacturer_Company_Id:** HDF-EQ-5521**

****Inventory_Quantity:** 3 units in stock**

****Supplier_Information:** Halliburton Houston Service Center, Contract #HDF-2023-3392, Lead Time: 10 days**

DOMAIN SPECIFICATIONS:

- ****Wellbore_Depth_Rating:** 15,000 feet TVD (matched to pump head rating)**
- ****Operating_Pressure_PSI:** 0 to 50 PSI suction side (vacuum rating: -12 PSI maximum)**
- ****Drilling_Fluid_Compatibility:** All mud types; 200-micron strainer element for standard operations**
- ****Known_Error_Codes:** ERR-SMA-001 (Strainer element clogging), ERR-SMA-002 (Check valve stiction), ERR-SMA-003 (Manifold crack/leak)**
- ****Replacement_Part_Number:** HDF-SMA-200-R1**
- ****Maintenance_Schedule:** Strainer element inspection every 250 hours; element replacement every 500 hours; manifold pressure test annually per ASME B16.5**

SECTION 1 – USE CASE

- Receives drilling fluid from mud tanks via suction line at atmospheric pressure
- Filters incoming mud through 200-micron strainer element to remove solids and protect pump internals
- Integral check valve prevents backflow and maintains prime during pump shutdown
- Provides pressure monitoring port for suction-side diagnostics and cavitation detection
- Supports mud tank circulation management and system redundancy

SECTION 2 – FAILURE CAUSES

- ****ERR-SMA-001 (Strainer Element Clogging)**** – Frequency: 41% of failures; excessive mud solids, inadequate tank cleaning, or high-viscosity mud systems
- ****ERR-SMA-002 (Check Valve Stiction)**** – Frequency: 31% of failures; mud residue buildup, corrosion of valve seat, or thermal cycling stress
- ****ERR-SMA-003 (Manifold Crack/Leak)**** – Frequency: 18% of failures; pressure surge events, thermal shock, or material fatigue at weld zones
- ****Suction Line Cavitation**** – Frequency: 7% of failures; excessive suction lift (>8 feet), undersized inlet line, or air leakage
- ****Strainer Bypass Valve Failure**** – Frequency: 3% of failures; spring fatigue or debris jamming

SECTION 3 – SOLUTIONS

- ****ERR-SMA-001 Mitigation:**** Replace strainer element per HDF-SMA-200-R1 specification; implement mud tank cleaning schedule (weekly minimum); upgrade to 150-micron element if solids

content exceeds 8% by volume per API RP 13B-1

- **ERR-SMA-002 Mitigation:** Disassemble check valve; clean valve seat and poppet with compatible solvent; replace if corrosion pitting exceeds 0.010"; verify spring preload (specification: 2.5 ± 0.3 PSI)
- **ERR-SMA-003 Mitigation:** Perform hydrostatic pressure test at 1.5× operating pressure per ASME B16.5; if leak detected, replace manifold assembly; inspect weld zones with dye penetrant testing (PT) per ASTM E1417
- **Suction Line Cavitation:** Verify suction lift ≤ 6 feet; increase inlet line diameter to minimum 4 inches; install vacuum breaker on tank vent; bleed air from suction line before operation
- **Strainer Bypass Valve Failure:** Replace bypass valve spring assembly; set cracking pressure to 3.5 PSI; perform functional test per manufacturer specification

SECTION 4 – INVENTORY DATA

- **Current Stock:** 3 complete manifold assemblies; 15 strainer elements (200-micron); 6 check valve cartridges; 4 bypass valve spring kits
- **Reorder Threshold:** Maintain minimum 2 manifold assemblies and 10 strainer elements; reorder when stock reaches 1 assembly and 5 elements
- **Supplier:** Halliburton Houston Service Center; secondary: Superior Drilling Components (Contract #SDC-2023-2156)
- **Lead Time:** 10 days standard; 5 days expedited; emergency 36-hour delivery available

UNIT 3: DISCHARGE MANIFOLD WITH RELIEF VALVE

Unit_Name: Discharge Manifold Assembly with Pilot-Operated Relief Valve

Unit_Id: PBDS-DMA-2847-A3

Manufacture_Date: 2023-05-20

Manufacturer_Company_Name: Baker Hughes Pressure Control Systems

Manufacturer_Company_Id: BHPCS-9847

Inventory_Quantity: 1 unit in stock

Supplier_Information: Baker Hughes Houston Technical Center, Contract #BHPCS-2023-5634, Lead Time: 12 days

DOMAIN SPECIFICATIONS:

- **Wellbore_Depth_Rating:** 15,000 feet TVD
- **Operating_Pressure_PSI:** 3,500 PSI nominal / 4,200 PSI relief setting (pilot-operated, adjustable ± 100 PSI)
- **Drilling_Fluid_Compatibility:** All mud types; rated for mud weights 8.5–18 ppg
- **Known_Error_Codes:** ERR-DMA-001 (Relief valve drift/creep), ERR-DMA-002 (Manifold internal erosion), ERR-DMA-003 (Pilot line blockage)
- **Replacement_Part_Number:** BHPCS-DMA-3500-R3
- **Maintenance_Schedule:** Relief valve setting verification every 250 hours; pilot line flushing every 500 hours; complete manifold overhaul every 6,000 hours

SECTION 1 – USE CASE

- Receives high-pressure discharge flow from pump head and directs to wellbore via standpipe and kelly
- Pilot-operated relief valve maintains system pressure within safe operating envelope (3,500–4,200 PSI)
- Provides pressure monitoring ports for surface readout and automated control system feedback
- Integrates with well control procedures and emergency shutdown systems per API RP 54 Section 8.1
- Supports mud return line backpressure management and equivalent circulating density (ECD) control

SECTION 2 – FAILURE CAUSES

- ****ERR-DMA-001 (Relief Valve Drift/Creep)**** – Frequency: 37% of failures; pilot line contamination, spring fatigue, or poppet seat erosion causing gradual pressure increase
- ****ERR-DMA-002 (Manifold Internal Erosion)**** – Frequency: 29% of failures; high-velocity mud impingement, abrasive particle erosion, or cavitation in discharge passages
- ****ERR-DMA-003 (Pilot Line Blockage)**** – Frequency: 18% of failures; mud solids accumulation in pilot line orifice, restricting control pressure signal
- ****Poppet Seat Galling**** – Frequency: 11% of failures; metal-to-metal contact during high-pressure cycling, inadequate lubrication
- ****Manifold Casting Porosity Leak**** – Frequency: 5% of failures; subsurface voids in ductile iron casting exposed under pressure cycling

SECTION 3 – SOLUTIONS

- ****ERR-DMA-001 Mitigation:**** Flush pilot line with clean mud at low pressure; replace pilot filter element (10-micron); recalibrate relief valve setting using calibrated pressure gauge per API RP 54 Appendix A; if drift persists, replace poppet and seat assembly per BHPCS-DMA-3500-R3
- ****ERR-DMA-002 Mitigation:**** Inspect discharge passages with borescope for erosion depth; if erosion exceeds 0.020", replace manifold; implement mud conditioning (centrifuge operation) to reduce particle size; reduce operating pressure by 5% during next 50 hours
- ****ERR-DMA-003 Mitigation:**** Disassemble pilot line; flush with compatible solvent; replace pilot filter element; verify pilot orifice diameter (specification: 0.040" ± 0.002"); perform functional test at 50% relief setting
- ****Poppet Seat Galling:**** Replace poppet and seat assembly; apply anti-seize compound (PTFE-based) to poppet stem; verify poppet travel clearance (0.015"–0.025")
- ****Manifold Casting Porosity Leak:**** Replace manifold assembly; perform 100% radiographic inspection (RT) per ASTM E1444 on replacement unit before installation

SECTION 4 – INVENTORY DATA

- ****Current Stock:**** 1 complete discharge manifold assembly; 3 relief valve cartridges; 6 poppet/seat kits; 8 pilot filter elements (10-micron)
- ****Reorder Threshold:**** Maintain minimum 1 complete manifold and 2 relief valve cartridges; trigger purchase order immediately when stock reaches threshold
- ****Supplier:**** Baker Hughes Houston Technical Center; secondary: Weatherford Pressure Control (Contract #WPC-2023-4821)

- **Lead Time:** 12 days standard; 6 days expedited (15% surcharge); emergency 48-hour delivery available for critical components

COMPLIANCE AND SAFETY SUMMARY

Regulatory Alignment:

- All procedures comply with OSHA 1910.119 (Process Safety Management) and OSHA 1910.147 (Lockout/Tagout)
- Equipment design and maintenance conform to API RP 54 (Recommended Practice for Occupational Safety and Health for Oil and Gas Well Drilling and Servicing)
- Pressure relief settings verified per API RP 520 (Sizing, Selection, and Installation of Pressure-Relieving Devices)
- Manifold pressure testing per ASME B16.5 (Pipe Flanges and Flanged Fittings)

Critical Safety Notes:

- All pressure system work requires LOTO procedures and pressure isolation verification
- Relief valve adjustments must be performed by certified technician with calibrated equipment
- Suction line cavitation prevention is critical to prevent pump cavitation damage and loss of prime
- Thermal management essential: discharge mud temperature must not exceed 180°F to prevent seal degradation
- Emergency shutdown procedures must be reviewed with all personnel before operation

Document Approval:

- Field Engineer: Marcus J. Whitmore (PBD-FE-4782)
- Date: 2024-01-18
- Next Review Date: 2024-07-18

END OF DOCUMENT